

STM32WL & LoRa/LoRaWAN Hands-on Training

1. STM32CubeIDE Essentials

- STM32CubeIDE installation & setup
 - Creating a new STM32 project
 - STM32CubeMX configuration
 - Project structure and code generation
 - Build, flash, and run your first application
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2. STM32CubeIDE Tips & Debugging

- Productivity tricks that save development time
 - ST-Link debugging
 - Breakpoints, Watch & Live Expressions
 - Register & Memory View
 - Debugging common firmware issues
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3. STM32WL & LoRa Fundamentals

- STM32WL architecture overview
 - LoRa vs LoRaWAN
 - RF parameters (SF, BW, CR, RSSI, SNR, TX Power)
 - Range optimization and antenna considerations
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4. (Optional) LoRa P2P Using AT Commands

Applicable only for new Seeed Studio Wio-E5-LE Mini boards with factory AT firmware

- Introduction to the Wio-E5 AT firmware
 - Basic AT commands
 - Configuring LoRa parameters
 - Sending & receiving P2P packets
 - Understanding packet parameters
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5. LoRa Development Using STM32CubeIDE

- Programming the STM32WL directly (replacing AT firmware)
 - Creating a LoRa project from scratch
 - Configuring the radio
 - P2P communication using the LoRa middleware
 - Custom payloads and application development
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6. LoRaWAN Fundamentals

- LoRaWAN architecture & data flow
 - Device Classes (A, B, C)
 - OTAA vs ABP
 - DevEUI, JoinEUI, AppKey & Session Keys
 - ADR, RSSI, SNR & Data Rate optimization
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7. Gateway & Network Server

- Gateway registration
- Device onboarding
- Frequency plan configuration

- OTAA activation
 - Monitoring uplinks & downlinks
 - Payload decoding
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8. LoRaWAN Stack Development

- Understanding the STM32WL LoRaWAN stack
 - Project structure
 - Adding sensors
 - Custom payload encoding
 - Handling downlinks
 - Building your own LoRaWAN application
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9. Low Power & Product Development

- Sleep modes and power optimization
 - Power profiling
 - Battery life estimation
 - Firmware architecture
 - Product development best practices
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10. Project Guidance & Mentorship

- Guidance on designing and developing your **Water Tank Monitoring Project**
- Architecture review and implementation approach
- Sensor integration and firmware development
- LoRa/LoRaWAN communication strategy
- Debugging support and best practices
- Recommendations for building a production-ready solution

Hands-on Projects

- LED Control over LoRa (P2P)
- **(Optional)** P2P Communication using AT Commands (Wio-E5 Factory Firmware)
- LoRa P2P using STM32CubeIDE
- LoRaWAN OTAA Join
- Uplink & Downlink Communication
- Custom Low-Power Sensor Node
- End-to-End LoRaWAN Application (Node → Gateway → Network Server → Dashboard)

Note: The primary focus of this training is **developing LoRa and LoRaWAN applications using STM32CubeIDE on the STM32WL platform**. The **AT Command session is optional** and is included only for participants using **new Seeed Studio Wio-E5-LE Mini boards with the factory-installed AT firmware**.